

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITDSA2-12

Group: 2

Lecturer's Name: Trudie Lotz

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Student Number: 4845350

Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature 	Date 18 March 2025
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Lecturer's Comments:

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Marks Awarded:	%
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Signature	Date
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Section A

Question 1

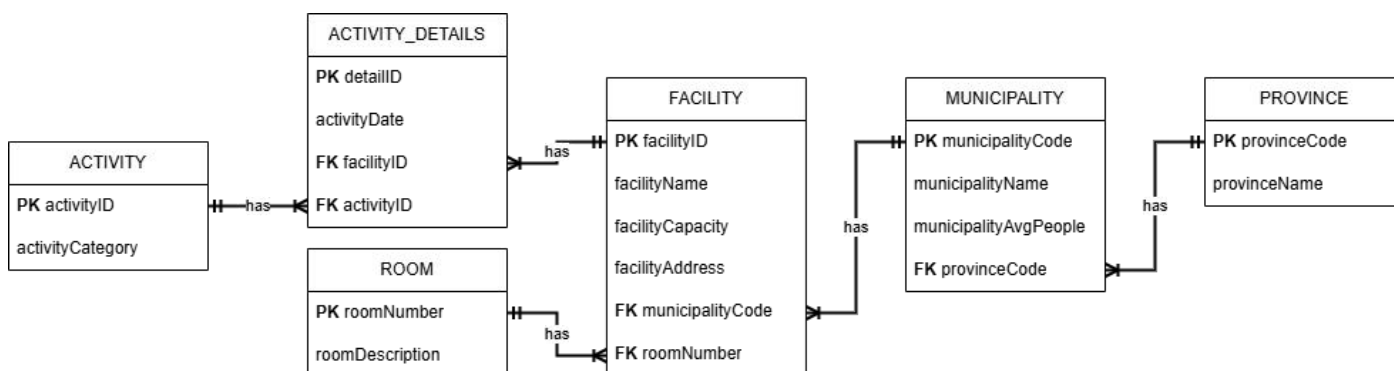
1.1)

Entity Name	Associated Attributes
FACILITY	facilityID, facilityName, facilityCapacity, facilityAddress
ROOM	roomNumber, roomDescription
MUNICIPALITY	municipalityCode, municipalityName, municipalityAvgPeople
PROVINCE	provinceCode, provinceName
ACTIVITY	activityID, activityCategory
ACTIVITY_DETAILS	detailID, activityDate

1.2)

Entity Name	Primary Key	Foreign Key
FACILITY	facilityID	municipalityCode, roomNumber
ROOM	roomNumber	NOTHING
MUNICIPALITY	municipalityCode	provinceCode
PROVINCE	provinceCode	NOTHING
ACTIVITY	activityID	NOTHING
ACTIVITY_DETAILS	detailID	activityCategory, facilityID

1.3)



Question 2

2.1)

Relationship entity pair	Relationship type
ACTIVITY – ACTIVITY_DETAILS	One-to-many, 1:M
ACTIVITY_DETAILS - FACILITY	One-to-many, 1:M
FACILITY - ROOM	One-to-many, 1:M
FACILITY - MUNICIPALITY	One-to-many, 1:M
MUNICIPALITY - PROVINCE	One-to-many, 1:M

2.2) RE-CHECK THIS!

0 NF	1 NF	2 NF	3 NF
facilityID facilityName facilityCapacity facilityAddress roomNumber roomDescription municipalityCode municipalityName municipalityAvgPeople provinceCode provinceName detailID activityID activityCategory activityDate	facilityID facilityName facilityCapacity facilityAddress roomNumber roomDescription municipalityCode municipalityName municipalityAvgPeople provinceCode provinceName detailID activityID activityCategory activityDate	facilityID facilityName facilityCapacity facilityAddress roomNumber roomDescription municipalityCode municipalityName municipalityAvgPeople provinceCode provinceName PK detailID activityDate FK facilityID FK activityID PK activityID activityCategory	PK facilityID facilityName facilityCapacity facilityAddress FK municipalityCode FK roomNumber PK roomNumber roomDescription PK municipalityCode municipalityName municipalityAvgPeople FK provinceCode PK provinceCode provinceName PK detailID activityDate FK facilityID FK activityID PK activityID activityCategory

2.3)

FACILITY						
	facilityID	facilityName	facilityCapacity	facilityAddress	municipalityCode	roomNumber
	NOT NULL	NOT NULL	NOT NULL	NOT NULL	NOT NULL	NOT NULL
Keys:	PK				FK	FK
Sample data:	f1	Apex Centre	150	Floor 1, East wing	m2	r1
	f2	Creativity Hub	165	Floor 2, Northern wing	m1	r2

ROOM		
	roomNumber	roomDescription
	NOT NULL	NOT NULL
Keys:	PK	
Sample data:	r1	Cinema
	r2	Theatre

MUNICIPALITY				
	municipalityCode	municipalityName	municipalityAvgPeople	provinceCode
	NOT NULL	NOT NULL	NOT NULL	NOT NULL
Keys:	PK			FK
Sample data:	m1	Cape Town Municipality	120	p1
	m2	Johannesburg Municipality	85	p2

PROVINCE		
	provinceCode	provinceName
	NOT NULL	NOT NULL
Keys:	PK	
Sample data:	p1	Western Cape
	p2	Johannesburg

ACTIVITY_DETAILS				
	detailID	activityDate	facilityID	activityID
	NOT NULL	NOT NULL	NOT NULL	NOT NULL
Keys:	PK		FK	FK
Sample data:	d1	20/02/2025	f1	a2
	d2	12/02/2025	f2	a1

ACTIVITY		
	activityID	activityCategory
	NOT NULL	NOT NULL
Keys:	PK	
Sample data:	a1	Circus
	a2	Music

Question 3

3.1)

- I. $\pi_{City, Phone} \left(\sigma_{Aname = "Mpho Sifiso"} (AUTHOR) \right)$
- II. $\pi_{Pname, Address, Phone} \left(\sigma_{City = "Cape Town"} (PUBLISHER) \right)$
- III. $\pi_{Book_title, Price} \left(\sigma_{Category = "Novel", Page_count > 200} (BOOK) \right)$
- IV. $\pi_{ISBN, Book_title, Price} \left(\sigma_{Category = "Novel", Category = "Textbook"} (BOOK) \right)$
- V. $\pi_{P_ID, Pname, Address, PhoneNo} \left(\sigma_{Category = "Novel"} \left(PUBLISHER \bowtie_{PUBLISHER.P_ID = BOOK.P_ID} BOOK \right) \right)$

3.2)

i)

Time	M	N
t1	read(Salary);	
t2	GrossSalary := Salary + Comm;	
t3	write(GrossSalary);	
t4	read(vatPercent);	
t5	NetSalary := GrossSalary – vatPercent;	
t6	write(NetSalary);	
t7	commit	
t8		read(hireData);
t9		YearsOfExperience := (CurrentDate – hireDate/365);
t10		CompensationPay := GrossSalary * YearsOfExperience;
t11		write(CompensationPay);
t12		commit

ii)

Time	M	N
t1	read(Salary);	
t2	GrossSalary := Salary + Comm;	
t3	write(GrossSalary);	
t4		read(hireData);
t5		YearsOfExperience := (CurrentDate – hireDate/365);
t6	read(vatPercent);	
t7	NetSalary := GrossSalary – vatPercent;	
t8	write(NetSalary);	
t9	commit	

t10		CompensationPay := GrossSalary * YearsOfExperience;
t11		write(CompensationPay);
t12		commit

3.3)

i)

Expression	Minimum size	Maximum size	Operation description
$R1 \times R2$	40,3	60,5	All possible pairs of values between R1 and R2 are selected.

ii)

Expression	Minimum size	Maximum size	Operation description
$R1 \cup R2$	3	60	All values in R1 and R2 that are not duplicates are selected.

iii)

Expression	Minimum size	Maximum size	Operation description
$R1 \cap R2$	Null	Null	All values that are in R1 and R2 are selected.

iv)

Expression	Minimum size	Maximum size	Operation description
$R1 - R2$	3	5	All tuples that are in R1 and not in R2 are selected.

v)

Expression	Minimum size	Maximum size	Operation description
$\sigma_{\text{Students} > 50}(R1)$	60	60	All values more than 50 from table R1 are selected.

vi)

Expression	Minimum size	Maximum size	Operation description
$\pi_{\text{Students}}(R1)$	40	60	All values from table R1 are projected.

vii)

Expression	Minimum size	Maximum size	Operation description
$R1 \bowtie R2$	3	60	Tables R1 and R2 are joined together with all the values.